$$\begin{cases} E = E_1 + E_2 \\ E_1 = \dfrac{1}{2} E_0 \cos(2\pi x / \lambda_x - 2\pi z / \lambda_z - \omega t) \\ E_2 = \dfrac{1}{2} E_0 \cos(2\pi x / \lambda_x + 2\pi z / \lambda_z - \omega t - \pi) \end{cases}, \tag{4}$$

Both waves have wavelength $\lambda_0$ given by

$$\left(\frac{1}{\lambda_0}\right)^2 = \left(\frac{1}{\lambda_x}\right)^2 + \left(\frac{1}{\lambda_z}\right)^2 = \left(\frac{\omega}{2\pi c}\right)^2 . \tag{5}$$

For $m$=1, $\lambda_z = 2d$ ,

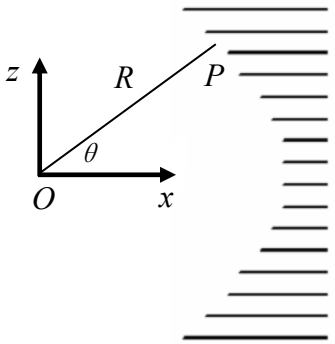
$$\lambda_x = \frac{1}{\sqrt{(2\pi c/\omega)^2 - (1/2d)^2}} . \tag{6}$$

3. Hence the wave speed along $x$, $v_x$ , is

$$v_x = (\omega / 2\pi)\lambda_x = \frac{c}{\sqrt{1 - \left(\dfrac{\pi c}{\omega d}\right)^2}} . \tag{7}$$

Any $P$ at the stack's left edge has source distance $OP$=$R$; $OP$ is at $\theta$ to horizontal. A horizontally propagating plane-wave output requires

$$L / c = R / c + (L - R\cos\theta) / v_x , \tag{8}$$




This gives

$$R = L\left(1 - \sqrt{1 - \left(\frac{\pi c}{\omega d}\right)^2}\right)\left(1 - \sqrt{1 - \left(\frac{\pi c}{\omega d}\right)^2}\cos\theta\right) . \tag{9}$$